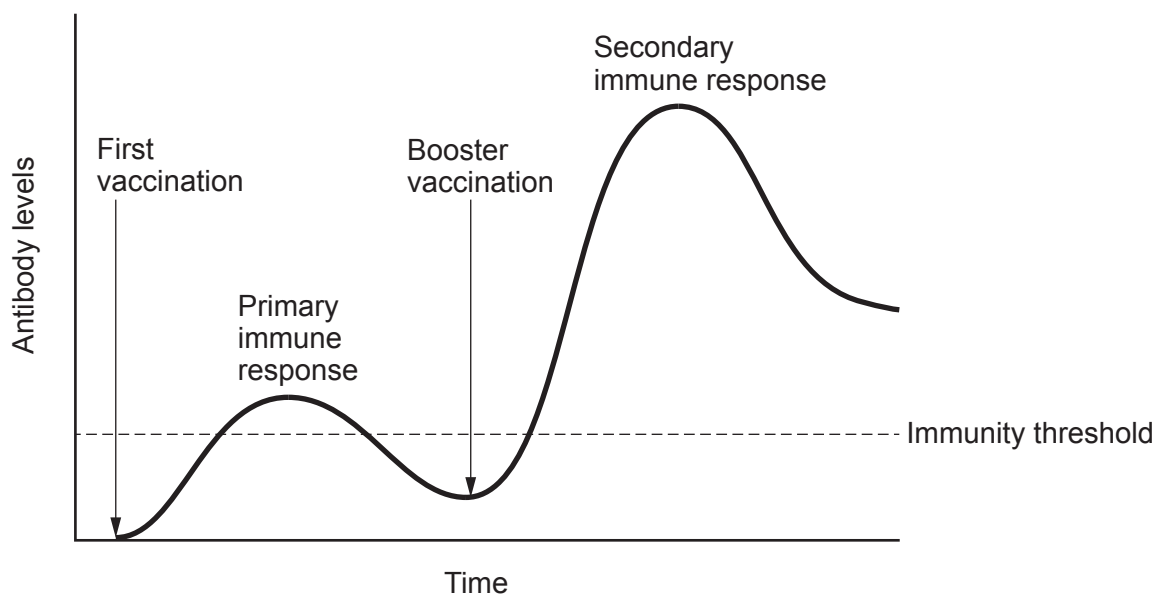


9. Vaccines protect against disease.

- (a) The graph below shows blood antibody levels following a first vaccination and a booster. Once antibody levels rise above a threshold value then there is immunity against the disease.



Compare the immune response following the initial vaccination and the booster vaccination.

[4]

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- (b) Explain how a vaccine protects against specific illnesses.

[3]

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THE PERIODIC TABLE

1 2 3 4 5 6 7 0

Group

1 H Hydrogen 1																	
7	9																
Li Lithium 3	Be Beryllium 4																
23	24																
Na Sodium 11	Mg Magnesium 12																
39	40	45	48	51	52	55	56	59	59	63.5	65	70	73	75	79	80	84
K Potassium 19	Ca Calcium 20	Sc Scandium 21	Ti Titanium 22	V Vanadium 23	Cr Chromium 24	Mn Manganese 25	Fe Iron 26	Co Cobalt 27	Ni Nickel 28	Cu Copper 29	Zn Zinc 30	Ga Gallium 31	Ge Germanium 32	As Arsenic 33	Se Selenium 34	Br Bromine 35	Kr Krypton 36
86	88	89	91	93	96	99	101	103	106	108	112	115	119	122	128	127	131
Rb Rubidium 37	Sr Strontium 38	Y Yttrium 39	Zr Zirconium 40	Nb Niobium 41	Mo Molybdenum 42	Tc Technetium 43	Ru Ruthenium 44	Rh Rhodium 45	Pd Palladium 46	Ag Silver 47	Cd Cadmium 48	In Indium 49	Sn Tin 50	Sb Antimony 51	Te Tellurium 52	I Iodine 53	Xe Xenon 54
133	137	139	179	181	184	186	190	192	195	197	201	204	207	209	210	210	222
Cs Caesium 55	Ba Barium 56	La Lanthanum 57	Hf Hafnium 72	Ta Tantalum 73	W Tungsten 74	Re Rhenium 75	Os Osmium 76	Ir Iridium 77	Pt Platinum 78	Au Gold 79	Hg Mercury 80	Tl Thallium 81	Pb Lead 82	Bi Bismuth 83	Po Polonium 84	At Astatine 85	Rn Radon 86
223	226	227															
Fr Francium 87	Ra Radium 88	Ac Actinium 89															

Key

A_r

Symbol

Name

Z

relative atomic mass

atomic number